Professional Installation Manual

Enterprise Server Deployment and Operating System Installation

Laguna State Polytechnic University
College of Computer Studies
ITEP 414 – System Administration and Maintenance
Week 3

Prepared By: Jasmin Clara S. Pacia

BSIT 4A_WAMD

Submitted To: John Randolf Peñaredondo

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1. Project Overview

Operating systems are the foundation of enterprise IT infrastructure. This Week 3 project assigns the student the role of a Junior System Administrator deploying a new Ubuntu Server for ABC Startup Solutions. The server will later support file sharing, remote administration, database hosting, web hosting, and internal services. The installation must follow company standards and be documented so another administrator can reproduce it.

2. Objectives

- Install Ubuntu Server in a virtual machine.
- Configure server settings during installation.
- Enable secure remote administration using SSH.
- Verify server functionality.
- Document installation procedures.
- Produce professional technical documentation.
- Differentiate BIOS and UEFI.
- Explain the computer boot process.
- Compare Windows Server, Ubuntu Server, and Rocky Linux.

3. Software Requirements

- Oracle VirtualBox or VMware Workstation.
- Ubuntu Server LTS ISO.
- Windows Server Evaluation ISO.

4. Virtual Machine Specifications

Component	Minimum Requirement
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 Virtual Processors
Storage	40 GB (VDI/VMDK)
Network	NAT (or Bridged if instructed)
Optical Drive	Ubuntu Server ISO

5. Ubuntu Server Installation Guide

1. Open Oracle VirtualBox or VMware Workstation.
2. Select New and create Ubuntu-Server-Week03.
3. Attach the Ubuntu Server ISO.
4. Allocate 4 GB RAM.
5. Assign 2 virtual processors.

6. Create a 40 GB VDI/VMDK virtual disk.
7. Configure networking as NAT unless Bridged is instructed.
8. Start the VM.
9. Select English.
10. Select the appropriate keyboard layout; use English (US) if applicable.
11. Allow network detection and use DHCP unless instructed otherwise. Record the assigned IP address.
12. Set hostname to server01.
13. Create a non-root administrative user using the required example fields.
14. Select Guided – Use Entire Disk and record partition scheme, file system, and disk size.
15. Select Install OpenSSH Server.
16. Do not install additional packages unless instructed.
17. Finish installation, restart the VM, remove the ISO if prompted, and log in.

6. Initial Server Configuration

Configuration Item	Required Value / Evidence
Hostname	server01
Administrative account	Non-root account
Network	DHCP unless otherwise instructed
IP address	Record assigned IP
Disk partition	Guided – Use Entire Disk; record scheme, file system, and size
SSH	OpenSSH Server installed
Additional packages	None unless instructed

7. Server Verification Procedures

Task 1 – Login

Log in using the account created and capture a screenshot.

Task 2 – Verify Hostname

hostname

Task 3 – Verify IP Address

ip addr

Task 4 – Test Internet Connectivity

ping -c 4 google.com

Task 5 – Update the Server

sudo apt update
sudo apt upgrade -y

Task 6 – Verify SSH Service
systemctl status ssh

The service should display active (running);.

7. Troubleshooting Guide

Problem / Symptom	Solution / Verification
VM does not boot from ISO	Confirm Ubuntu Server ISO is attached to the optical drive and the VM configuration matches the required Week 3 specifications.
No IP address	Check that the network adapter is enabled and configured for NAT (or the instructed network mode), then verify using `ip addr`.
Hostname is wrong	Run `hostname` and correct the hostname if it does not show `server01`.
Internet connectivity fails	Check the IP address and VM network adapter, then repeat `ping -c 4 google.com`.
APT update/upgrade fails	Confirm network connectivity, then rerun `sudo apt update` followed by `sudo apt upgrade -y`.
SSH is not active	Run `systemctl status ssh` and verify that Install OpenSSH Server was selected during installation.
Login fails	Use the non-root administrative account created during installation and verify the credentials.
ISO remains mounted after installation	Restart the VM and remove the installation ISO if prompted.

8. Best Practices

- Follow the required VM specifications.
- Use a non-root administrative account.
- Record the assigned IP address.
- Use DHCP unless instructed otherwise.
- Verify OpenSSH after installation.
- Run the required update commands.
- Capture all required screenshots.
- Record actual problems and solutions.
- Organize screenshots/ and diagrams/ correctly.
- Do not install additional packages unless instructed.